swipe →

Empty shelves lose sales.

Full storerooms lock up cash.

I built an ML system that predicts demand and tells a store what to order, and when.

−67%	−40%	−30%
stockout days	inventory cost	forecast error

The problem

A 3-store supermarket reorders with a rule of thumb:

> *“When stock looks low,*
> *order 3 weeks of average sales.”*

It ignores growth, seasons and Diwali. Over 2 years:

7.9% of store-product days were out of stock

Rs. 1.53 Cr of revenue walked out of the door

26% stockout days for Electronics

How it works

One command. Seven steps.

1 **Simulate**
2 yrs × 3 stores × 20 products: trend, seasons, festivals, promos, supplier delays

2 **Clean**
duplicates, outliers, missing values → 0 missing

3 **Explore**
8 charts: weekday, seasonality, promo uplift

4 **Engineer**
17 leak-free features + festival-free baseline

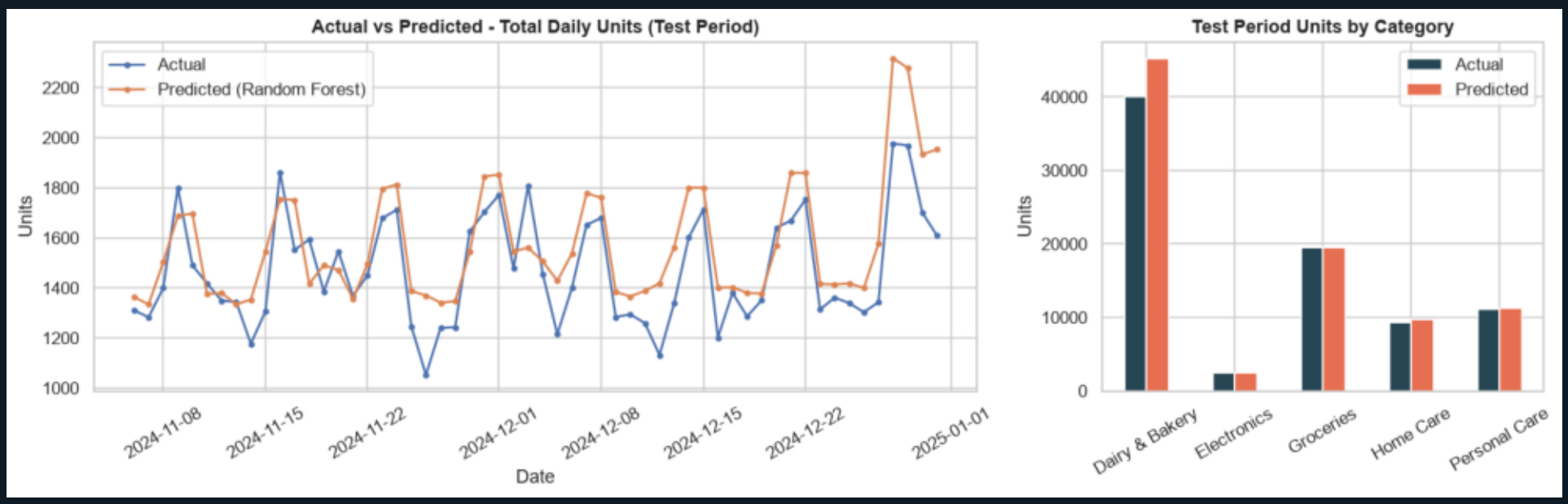
5 **Forecast**
Baseline vs Linear Regression vs Random Forest

6 **Optimize**
Safety stock · Reorder point · EOQ

7 **Simulate impact**
old rule vs new policy on real demand

Forecasting

Random Forest beats the baseline by 30%



	daily WAPE	weekly	bias
Seasonal Naive (baseline)	54.2%	35.0%	+31%
Linear Regression	39.9%	21.3%	+8%
Random Forest	38.1%	20.4%	+7%

Tested honestly: 56 days the model never saw, feeding its own predictions forward like real life.

From forecast to decision

Safety stock	= 1.645 × σ × √lead time
Reorder point	= lead-time demand + safety stock
Order quantity	= max(EOQ, target − stock)

Every morning: a list of what to order, and how much

Reorder Alerts - Action Required Today

Store	Product	Stock	ROP	Safety	Order Qty	Status
S01	Shampoo 340ml	0	122	32	394	CRITICAL
S01	LED Bulb 9W	0	134	49	446	CRITICAL
S01	Bluetooth Earbuds	0	48	15	67	CRITICAL
S01	Electric Kettle	0	66	21	110	CRITICAL
S02	Detergent Powder 2kg	0	108	30	311	CRITICAL
S02	Bluetooth Earbuds	0	32	10	56	CRITICAL
S03	Room Freshener	0	57	16	294	CRITICAL
S03	Bluetooth Earbuds	0	28	10	56	CRITICAL
S03	Electric Kettle	0	34	13	60	CRITICAL
S02	Electric Kettle	1	33	16	57	CRITICAL
S01	Basmati Rice 5kg	29	134	37	325	CRITICAL
S01	Wheat Atta 10kg	57	97	32	359	REORDER

The impact

Same demand. Same starting stock. Old rule vs my policy:

	OLD	NEW
Fill rate	97.8%	**99.2%**
Stockout days	6.7%	**2.2%**
Lost revenue	₹9.2L	**₹2.7L**
Purchase orders	203	**162**
Total cost	₹3.68L	**₹2.22L**

Trade-off: 31% more average stock, held where demand needs it.
(Simulated data, real methods.)

Built with

- Python
- Pandas
- NumPy
- scikit-learn
- Matplotlib
- Seaborn
- Streamlit
- pytest

What I learned

- Time-based validation & data leakage
- Why bias matters as much as accuracy
- Safety stock, reorder point & EOQ
- Turning model metrics into ₹ impact

Full code + dashboard on GitHub · link in bio